Meeting Leech Detector: Project Progress and New Features

This document explains the major changes and new features added to the **Meeting Leech Detector (MLD)** since the first version of the project (Minor Project 1).

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1. Why Did We Make Changes?

In the first version of the project, the system relied on tracking employees using their **IP Addresses** (a numerical label assigned to a device connected to a computer network).
While this worked for basic testing, it had major real-world problems:

- **Working from Home:** If an employee's home internet restarted, their IP address changed, and the system lost track of them.
- **Shared Networks:** If multiple employees worked from the same office or coffee shop, they all shared the same public IP address, causing confusion.
- **VPNs:** Many corporate employees use VPNs (Virtual Private Networks) which hide their real IP addresses entirely.

To solve this, we completely removed IP-address tracking and built a much more flexible and secure system.

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2. New Features Added

A. The "Session Code" System (Replaces IP Addresses)

Instead of relying on network addresses, we built a **Session Code** system (similar to joining a Kahoot game or a Zoom meeting via a link).

- **How it works:** The Manager clicks a button to start a meeting and the system generates a unique 6-character code (e.g., `MLD123`). The manager shares this code with the employees.
- **Why it's better:** Employees just enter the code into their dashboard to join the monitoring session. They can be anywhere in the world, on any network, and the system will perfectly track them.

B. Webcam Activity Checking

We wanted to ensure employees actually have their cameras turned on during important meetings, without invading their privacy by recording video.

- **How it works:** The system now securely checks the computer's internal settings (the Windows Registry) to see if the webcam hardware is currently broadcasting. It doesn't look at the video; it just asks the computer, *"Is the camera on?"*

C. Upgraded Database Engine

- **The Change:** We upgraded the underlying database from a simple, single-file system (SQLite) to a robust, professional-grade database engine (**PostgreSQL**).
- **Why it matters:** This allows the backend to handle hundreds of employees sending data at the exact same time without slowing down or crashing.

D. Modern "Glassmorphism" Design

- We completely redesigned the visual look of the application for both Employees and Managers. It now features a premium, modern design with semi-transparent elements (glassmorphism), smooth animations, and real-time updating numbers that make the software feel incredibly polished and professional.